- It is essential for the metabolism and proteins, fats and carbohydrates.
- It assists in the maintenance of serum level of sodium and potassium
- It helps to promote red blood cell production.
- It is linked to cancer immunity and fights the formation of the toxic chemical homocysteine.

Food sources

Good sources to obtain pyridoxine are brewer's yeast, egg, chicken, carrot, fish, liver, kidney, pea, wheat germ and walnuts. Roots and tubers, cabbage, legumes, molasses, whole grains, etc., contain moderate amount of this vitamin.

Deficiency

Irritability, nervousness, insomnia, anemia, general weakness, skin changes such as dermatitis.

Folic acid (Vitamin B9) – folic acid, folacin, folate

Folic acid is also referred to as folacin or folate. Its chemical name is pteroylglutamic acid. This vitamin can be produced by the body and be stored in the liver.

Functions

- Folic acid is required for DNA synthesis and cell growth and is important for red blood cell formation, energy production as well as the forming of amino acids.
- Folic acid is essential for synthesizing heme, the iron containing substance in hemoglobin, crucial for oxygen transport.

Folic acid is very important in the development of the nervous system of a developing fetus.

Food sources

Fresh green vegetable such as spinach and broccoli contains folic acid. It is also found in fruit, starchy vegetables, beans, whole grains, liver, kidney, egg, yeast etc.

Deficiency

A deficiency of folate can lead to macrocytic, megaloblastic anemia, diarrhea, fatigue, depression, and mental confusion.

Cyanocobalamin (Vitamin B12)

Cyanocobalamin also known as cobalamin is referred to as the energy vitamin. It is a very widely researched vitamin, and used in supplementation to a very large degree.

Functions

- Cobalamin is required in the metabolism of fats, proteins and carbohydrates.
- It is needed in the manufacture of red blood cells and the maintenance of red blood cells.
- It stimulates appetite and Promotes growth.

Food sources

Liver, organ meat, muscle meat, shellfish, egg, cheese and fish are rich sources of this vitamin. It can be manufactured in the body. Milk contains vitamin B_{12} however processing of milk may destroy the vitamin.

Deficiency

Vitamin B12 deficiency results in macrocytic, megaloblastic anemia (pernicious anemia) similar to that occurring with folate deficiency.

Biotin (Vitamin B7)

Biotin is also referred to as anti – egg white injury factor.

Biotin present in foods in not affected by exposure to light. Biotin is also produced by bacteria in the intestine.

Function

- Biotin is involved in carbohydrate and lipid metabolism.

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